

DAY 2

Forest type and carbon

Make your own map instead of using someone else's, then put a number on what it stores.

2 hours · Classification + Carbon

Labs 3 and 4

RECAP

Where we got to

Yesterday you learned

- An image is a grid of measurements, in many bands
- NIR is what makes vegetation mapping possible
- NDVI turns bands into one number
- The median beats clouds
- **Always check the number**

THREE QUICK QUESTIONS

1. Your NDVI map has a maximum of 2.4. What went wrong?
2. Why did we use `scale=100` and not `scale=10`?
3. WorldCover and Dynamic World disagreed by 52,000 ha. Which one is correct?

Answers: (1) values were not divided by 10000 — NDVI cannot exceed 1. (2) A hundredth of the computing for the same answer. (3) **Neither.** They measure slightly different things, and both must be cited with their name and year.

PART ONE

Classifying forest types

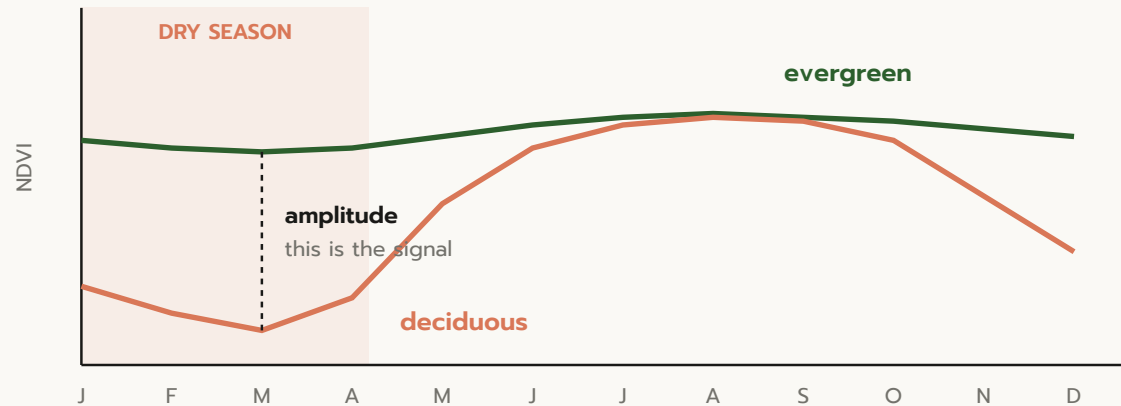
The satellite cannot see species. But it can see behaviour.

You cannot tell evergreen from deciduous in one image.

In the wet season they are both simply green. The information is not in the picture — it is in the **change between pictures**.

THE PHYSICAL IDEA

Deciduous forest gives itself away in the dry season



Evergreen forest holds its leaves all year, so its NDVI barely moves.

Deciduous forest drops its leaves when the rain stops, and its NDVI collapses.

THE WHOLE METHOD

$$\text{amplitude} = \text{NDVI}_{\text{wet}} - \text{NDVI}_{\text{dry}}$$

Low amplitude → evergreen.

High amplitude → deciduous.

CHECK FIRST

Is the signal actually there?

Before trusting an idea, **measure it**. If the two forest types really differ, amplitude values over forest should be spread out — not clustered at one value.

So we take the percentiles of NDVI amplitude across everything WorldCover calls tree cover:

The middle half of the forest spans **0.07 to 0.26**. That is a wide spread, so there really are two different kinds of forest here.

Had every percentile sat within 0.02 of the median, the honest move would be to **abandon this approach** — not to push on and produce a confident-looking map of noise.

NDVI AMPLITUDE OVER TREE COVER · NAN

5th percentile	-0.019
25th	0.074
median	0.160
75th	0.262
95th	0.394

Random Forest, in one slide

A **Random Forest** is a few hundred simple decision trees, each trained on a random slice of your data, all voting on the answer.

It has nothing to do with real forests. The name is an unfortunate coincidence that has confused foresters for twenty years.

You need three things:

- **Labelled examples** — "this pixel is deciduous"
- **Predictors** — the bands and indices it may look at
- **Held-back data** — to find out whether it worked

THE ONE RULE YOU MUST NOT BREAK

Never measure accuracy on the points you trained with.

The model has already seen them. It will score beautifully and tell you nothing.

Split 70 / 30. Train on the 70. Judge on the 30 it has never met. If your accuracy comes back **0.99**, be suspicious rather than pleased — that is what a leak looks like.

Results you should get

• LIVE — Lab 3

0.863

overall accuracy

0.816

kappa

Good, and believably imperfect. A realistic range for this kind of work is **0.75 to 0.92**.

NAN PROVINCE · CLASSIFIED AREA

Deciduous forest	352,246 ha
Evergreen forest	332,599 ha
Cropland	152,866 ha
Water / built	29,092 ha

Northern Thailand genuinely is a mixed deciduous landscape, so a near 50/50 split is the right kind of answer. 95% evergreen would mean your thresholds need adjusting.

The confusion matrix tells you where the map fails

	EVERG.	DECID.	CROP	WATER
Evergreen	79	0	2	2
Deciduous	0	73	8	2
Cropland	4	10	51	8
Water/built	0	1	10	92

Rows are truth, columns are what the model said.

Cropland is the weak class. Ten cropland points were called deciduous forest.

That is not a bug — it is physics. In April a harvested field and a leafless forest look genuinely similar from orbit.

If you present this map, **say this out loud.** "Overall accuracy 86%, with cropland and deciduous forest the main confusion" is a professional sentence. "86% accurate" on its own is not.

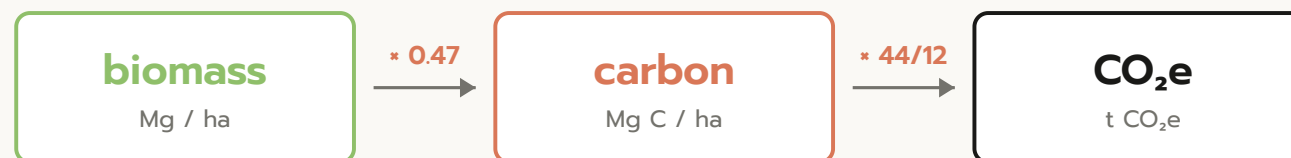
PART TWO

Putting a number on carbon

Where forest carbon figures come from, and how far apart they can be.

THE CHAIN

From tree to tonne of CO₂



0.47 – THE CARBON FRACTION

About 47% of dry wood is carbon. The IPCC default, used almost universally.

44/12 = 3.667

One tonne of carbon, fully oxidised, becomes 3.667 tonnes of CO₂ – the mass of the oxygen it picks up.

Every carbon figure you will ever read walks this chain. **The mistakes happen at the first arrow** – because some datasets have already applied it and some have not.

Two datasets. Both called "aboveground biomass". Different units.

One has already done the $\times 0.47$. The other has not. Multiply the wrong one and your answer is off by a factor of two — with no error message.

THE TRAP

Read the units, every single time

DATASET	BAND	UNITS	WHAT YOU MUST DO
NASA/ORNL/biomass_carbon_density/v1	agb	Mg C/ha already carbon	Do NOT multiply by 0.47.
LARSE/GEDI/GEDI04_B_002	MU	Mg/ha biomass	DO multiply by 0.47.

AND A SECOND TRAP IN THE SAME DATASET

```
# what every AI writes – and it fails
ee.Image('NASA/ORNL/biomass_carbon_density/v1')

# what actually works
ee.ImageCollection('NASA/ORNL/...').first()
```

It is an **ImageCollection** containing exactly one image. The catalog page does not shout about it.

We tested this on Claude, Gemini and ChatGPT. **All three got it wrong.** This is what the cheat sheet is for.

Nan province carbon

• LIVE — Lab 4

63.9_M

tonnes of carbon
in Nan forest

234_M

tonnes CO₂ equivalent

If you got thousands, you forgot to multiply by pixel area. If you got trillions, you used m² instead of hectares.

CARBON DENSITY · NAN FOREST · ORNL

Aboveground	46.2 Mg C/ha
Belowground	13.4 Mg C/ha
Total	59.5 Mg C/ha

Tropical forest carbon runs roughly 40–200 Mg C/ha, so this is plausible for a mixed deciduous landscape.

Now check it against a second dataset. It disagrees by 67%.

ORNL says 46 Mg C/ha. GEDI says 163 Mg/ha of biomass — which is 77 Mg C/ha. Both are peer-reviewed and widely cited.

Three real reasons, none of them a bug

- **Different years.** ORNL is a 2010 snapshot. GEDI flew 2019–2021. Nine years of growth and loss sit between them.
- **Different instruments.** GEDI is a spaceborne **lidar** measuring canopy height directly. ORNL blends earlier maps.
- **Different coverage.** GEDI only reports where its laser landed — about **60%** of this province.

SO REPORT IT LIKE THIS

*"Aboveground carbon in Nan forest is estimated at **46–77 Mg C/ha** (ORNL 2010; GEDI L4B 2019–2021)."*

A **range with sources** is honest and defensible. A single number to three decimal places is neither — and any reviewer who knows the field will ask you which product you used.

Day 2 wrap

What you can now do

- Separate evergreen from deciduous using seasonal behaviour
- Train a Random Forest and measure it honestly
- Read a confusion matrix, not just an accuracy score
- Convert biomass to carbon to CO₂e — in the right direction
- Report a range with its sources

HOMEWORK — 15 MINUTES

Run Lab 3 on your own province and write down the accuracy.

Then change `numPoints` from 300 to 50 and run it again. **Note what happens to the accuracy** — and come tomorrow able to say why.

Tomorrow: fire and climate — and a map that looks completely convincing while being wrong by a factor of twenty-four.